



Indian Institute of Technology Kanpur  
Department of Mathematics and Statistics  
**Introduction to Bayesian Analysis (MTH422)**  
Quiz 3, Date: March 24, 2025, Monday

Time: 4:00–4:40 PM (4:00–4:55 PM for DAP)

Spring 2025

Max point: 10

B

**There will NOT be any partial credit in any question.** All assumptions (whether observations are IID or not, independent or not) should be explicitly and correctly written.

The following is a JAGS snippet and outputs. Answer the questions on the next page based on these codes and outputs.

```
1 data <- list(Y = Y, n = length(Y))
2 model_string <- textConnection("model{
3 for(i in 1:n){
4 Y[i] ~ dnorm(X[i], 400)
5 }
6 for(i in 1:n){
7 X[i] ~ dnorm(mu, tau)
8 }
9 tau ~ dgamma(0.001, 0.001)
10 sigma <- 1/sqrt(tau)
11 mu ~ dnorm(0, 0.01)
12 }")
13
14 model <- jags.model(model_string, data=data, n.chains=1, quiet=TRUE)
15 update(model, 20000, progress.bar = "none")
16
17 params <- c("mu", "sigma")
18 samples <- coda.samples(model, variable.names = params,
19                          n.iter = 40000, thin = 4, progress.bar = "none")
20 summary(samples)
21
22 1. Empirical mean and standard deviation for each variable,
23    plus standard error of the mean:
24
25      Mean      SD Naive SE Time-series SE
26 mu    1.9665 0.02767 0.0002767      0.0002767
27 sigma 0.2715 0.02063 0.0002063      0.0002021
28
29 2. Quantiles for each variable:
30
31      2.5%   25%   50%   75%   97.5%
32 mu    1.9116 1.948 1.9662 1.9849 2.0210
33 sigma 0.2344 0.257 0.2704 0.2847 0.3147
34
35 geweke.diag(samples) # based on default setting
36      mu      sigma
37 -0.4128  0.4076
```

1. Write down the two layers of the random effect model and the two priors. All parameters and hyperparameters (parameters of the prior), independence or IID-type conditions should be explicit. ( $0.5 \times 4 = 2$  points)
2. What are the burn-in length, post-burn-in length before thinning, post-burn-in length after thinning? How many samples are used for drawing inference here? ( $0.5 \times 4 = 2$  points)
3. What is the dimension of the posterior distribution in this problem? Write down the purpose of Line 17 of the snippet in this context. ( $0.5 \times 2 = 1$  point)
4. Calculate the effective sample size for  $\sigma$ . (1 point)
5. Report the Monte Carlo estimates of the 95% equal-tailed posterior credible interval of  $\mu$  and  $\sigma$ . ( $0.5 \times 2 = 1$  point)
6. The Monte Carlo estimate of the posterior probability of  $\sigma$  being within the interval (0.25, 0.3) is at least 0.5. Justify. (1 point)
7. Instead of the posterior of  $\sigma$ , if our interest lies in the posterior of  $\sigma^2$ , how would you modify the above code? Further, if you want to obtain the inference about  $\mu + 2\sigma$ , how would you modify the above code? ( $0.5 \times 2 = 1$  point)
8. What can you conclude from the Geweke statistics reported here? Calculate the Gelman-Rubin diagnostic here. If you cannot calculate, justify why. ( $0.5 \times 2 = 1$  point)